OFFICIAL USER MANUAL

BARBELL LABS

Everything you need to log smarter, train harder, and track every PR.

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01 🚀 GETTING STARTED

Barbell Labs is a workout tracker built for everyone who trains — gym, home, garage, or anywhere you break a sweat. Getting started takes less than 60 seconds.

CREATING YOUR ACCOUNT

1 Open the app and tap "Create Account"

Enter a username, your email address, and a password (min. 8 characters, 1 uppercase, 1 lowercase, 1 number).

2 Verify your email

A verification link is sent to your inbox. Tap it to confirm your account. Check spam if you don't see it.

3 Sign in and set up your profile

Go to the Profile tab → tap Edit to add your name, age, weight, height, city, and training goal.

4 Start logging

Tap the Log tab, hit "Add Exercise", pick from the 224-exercise library, and record your first set.

> 💡 TIP
> You can also sign in with Google for one-tap account creation — no password required.

NAVIGATION

TAB	WHAT IT DOES
🏠 Home	Overview — greeting, streak, quick stats, recent sessions, quick-log button
📋 Log	Active workout logger — add exercises, record sets, use the rest timer
📖 History	All past workouts — view, search, expand, export to CSV
📈 Progress	Charts, PR tracking, muscle breakdown, exercise progression graphs
👤 Profile	Personal info, goals, lifetime stats, settings, account security

02 📋 LOGGING A WORKOUT

The Log tab is where you record every set, rep, and weight. Everything you log today becomes the coaching and charts you see tomorrow.

STARTING A WORKOUT

1 Tap the Log tab

Or press "Start Workout" on the Home tab to jump straight into logging.

2 Tap "Add Exercise"

Opens the exercise picker. Search by name, or filter by muscle group and equipment type.

3 Log your sets

Enter weight and reps for each set. Tap "+ Add Set" to add more. Tap the trash icon to remove a set.

4 Tap "Finish Workout" when done

The app saves your session, calculates duration, updates your streak, and shows your post-workout summary.

SET FIELDS

FIELD	DESCRIPTION
Weight	The load used. Enter 0 or leave blank for bodyweight exercises.
Reps	The number of reps completed in that set.
RPE	Rate of Perceived Exertion (1–10). How hard the set felt. Tap to toggle.
RIR	Reps in Reserve. How many more reps you had left. Tap to toggle.
Notes	Freeform text for any set — form cues, how you felt, variations used.

WORKOUT LABELS

Tag each workout with up to 3 labels (Legs, Push, Pull, Upper Body, Lower Body, Shoulders, Arms, Abs, Glutes). Labels appear in History to help you spot patterns.

> **QUICK LOG**
>
> On the Home tab, tap "Quick Log Last Session" to instantly reload your most recent workout as a starting point. Edit any sets you want to change, then finish.

03 📚 EXERCISE LIBRARY

Barbell Labs includes a complete library of **224 exercises** across 8 muscle groups and 6 equipment types — the full Gym Bible, built in.

FILTERING EXERCISES

When you open the exercise picker, you'll see two rows of filter chips above the results list.

💪 Muscle Group (swipe left/right)

All · Chest · Back · Shoulders · Arms · Legs · Core · Cardio · Full Body

🏋 Equipment (swipe left/right)

All Equip · Barbell · Dumbbell · Machine · Cable · Bodyweight · Other

Use the search bar to find any exercise by name. If an exercise isn't in the library, type the name and tap **+ Add "[name]"** to create a custom entry.

MUSCLE GROUPS

Chest
27 exercises

Back
30 exercises

Shoulders
21 exercises

Arms
27 exercises

Legs
37 exercises

Core
28 exercises

Cardio
19 exercises

Full Body
19 exercises

04 🤖 AI COACH

After every set, Barbell Labs analyses your entire history for that exercise and shows you a personalised recommendation — exactly what weight and reps to aim for next.

HOW IT WORKS

📈 Progressive Overload

If you've been consistent and RPE was moderate, the coach nudges up weight or reps to drive adaptation.

🔄 Plateau Detection

Stalled at the same weight for 3+ sessions? The coach tells you it's time to break through — or suggests a deload.

🩹 Recovery Mode

Coming back after time off? The coach eases you back in with a reduced target to protect against injury.

💪 RIR & RPE Aware

Logged high RIR (lots in the tank)? The coach pushes you harder. Logged RPE 9–10? It may suggest backing off slightly.

> 💡 TIP
> The more sessions you log, the smarter the coach becomes. After 3–5 sessions on an exercise you'll start seeing highly accurate recommendations.

USING SUGGESTIONS

The AI recommendation appears as a banner above the set rows. Tap **Apply** to pre-fill the weight and reps into your next set, or **Dismiss** to ignore it for this session.

05 ⏱ REST TIMER

The built-in rest timer sits inside every exercise block. It keeps your rest periods consistent and signals you when it's time to go again.

USING THE TIMER

1 Choose a preset

Tap 30s, 1m, 2m, 3m, 5m — or tap Custom to enter any minutes:seconds combination.

2 Tap Start

The timer counts down. You'll feel a haptic vibration when it hits zero.

3 Resume or Reset

Tap Pause to stop mid-rest. Tap Reset to cancel and start fresh.

🔔 **HAPTIC FEEDBACK**

When the timer reaches zero, your phone vibrates to alert you even if your screen is off or you're mid-conversation.

06 🎯 RPE & RIR TRACKING

Track effort, not just weight. RPE and RIR give you and the AI Coach the full picture of how hard each set really was.

METRIC	SCALE	WHAT IT MEANS
RPE Rate of Perceived Exertion	1–10	How hard the set felt. 10 = absolute max effort, no reps left. 7 = moderately hard, a few left.

RIR Reps in Reserve	0–5+	How many more reps you could have done. 0 = failure. 3 = three reps left in the tank.

RPE QUICK REFERENCE

RPE	FEEL	RIR EQUIV.
10	Max effort — couldn't do another rep	0
9	Could squeeze out 1 more rep	1
8	Hard — 2 reps left	2
7	Moderate — 3 reps left	3
6	Comfortable — could go much harder	4+

💡 **TIP**

Log RPE and RIR consistently and the AI Coach will use that data to make progressively smarter recommendations.

07 📈 PROGRESS & CHARTS

The Progress tab shows dual-line charts for every exercise you've logged — weight and reps plotted together over your entire history.

READING YOUR CHARTS

📊 Dual-Line Chart

Each exercise graph shows two lines: **weight** (solid, steel blue) and **reps** (dashed, lighter). Both rising together = ideal progression.

🔍 Scrubbing

Drag your finger along the chart to scrub through history session by session. The tooltip shows exact weight, reps, and date for each point.

👑 1RM Estimate Badge

Each exercise card shows your estimated one-rep max (Epley + Brzycki average). Tap the "1RM" button in Log to get a full percentage table.

> 💡 **THE LONGER YOU LOG, THE MORE POWERFUL IT GETS.**
> After 10+ sessions per exercise, trends and plateaus become clearly visible.

08 🏆 MY TOP LIFTS

The top of the Progress tab shows your personal records for your three most important exercises. By default these are Barbell Bench Press, Barbell Back Squat, and Conventional Deadlift — but you can change them to any exercise in the library.

CUSTOMISING YOUR TOP LIFTS

1 **Tap "Edit" next to My Top Lifts**

The three lift slots expand into individual pickers.

2 **Tap any slot to change it**

A search field appears. Type to filter all 224 exercises. Tap one to select it.

3 **Tap "Save"**

Your choices are saved to your profile and synced across devices.

> 👑 **CROWN BADGE**
>
> The lift with your highest PR across all three receives a "TOP PR" crown badge. Chase it every session.

09 💯 1RM ESTIMATOR

Your estimated one-rep max tells you your theoretical maximum for a lift based on what you actually lifted, without ever having to max out.

ACCESSING THE 1RM CALCULATOR

In the Log tab, tap the **1RM** pill button in the top-right corner. Enter any weight and rep count to instantly see your estimated 1RM.

PERCENTAGE TABLE

Below your 1RM estimate, a full percentage table shows training loads from 50% to 100% — helping you program exactly the right weight for any rep range.

% OF 1RM	REP RANGE	TRAINING ZONE

90–100%	1–3 reps	Max strength / peaking
75–89%	4–8 reps	Strength + hypertrophy
60–74%	8–15 reps	Hypertrophy
50–59%	15+ reps	Endurance / warmup

🔢 **FORMULA**

Uses the average of the Epley and Brzycki formulas for maximum accuracy across all rep ranges.

10 🏋️ PLATE CALCULATOR

Know exactly which plates to load before you even walk up to the bar. Tap **Plates** in the Log tab header.

HOW TO USE

1. **Select your unit (lbs or kg)**
   Toggle between imperial and metric with the unit switch.

2. **Choose your bar weight**
   Standard 45lb / 20kg, Women's 35lb / 15kg, or EZ-Bar 25lb / 11kg.

3. **Enter your target weight**
   The calculator instantly shows which plates go on each side, with a colour-coded visual bar diagram.

🎨 **COLOUR CODING**

Each plate size has its own colour matching standard gym plate colours — red 25s, blue 20s, yellow 15s, green 10s, white 5s, and so on.

11 📄 WORKOUT TEMPLATES

Save any completed workout as a template and reload it in one tap on any future session.

SAVING A TEMPLATE

1 Complete a workout

On the post-workout summary screen, tap "Save as Template".

2 Name your template

Give it a memorable name — "Push Day A", "Leg Day", "Upper Body Volume", etc.

3 Tap Save

The template is stored in your account and synced to all devices.

LOADING A TEMPLATE

In the Log tab, tap **Templates** to open your saved list. Tap any template to load it as your current workout — all exercises and set structure pre-filled. You just enter the weights and reps.

MANAGING TEMPLATES

Open the Templates sheet to rename or delete any template. Changes sync immediately.

12 💪 MUSCLE GROUP BREAKDOWN

See exactly which muscle groups you've been training — and which ones you might be neglecting — over the last 7 days, 30 days, or full year.

READING THE BREAKDOWN

📊 Proportional Bar Chart

Each muscle group shows a bar sized proportionally to the number of sets you logged. Bigger bar = more volume that muscle received.

🗓 Time Ranges

Toggle between **7D** (last 7 days), **30D** (last 30 days), and **1Y** (last year) to spot weekly imbalances vs. long-term patterns.

> 💡 **USE THIS TO CATCH IMBALANCES EARLY.**
> If your push muscles (chest, shoulders, triceps) are dominating over pull muscles (back, biceps), adjust your programming before it causes issues.

13 📖 WORKOUT HISTORY

Every workout you've ever completed lives in the History tab — fully searchable, expandable, and exportable.

BROWSING HISTORY

Sessions are listed newest first. Tap any session card to expand the full exercise list with every set, weight, and rep. Workout labels appear as coloured badges on each card.

EXPORTING TO CSV

Tap the **Export CSV** button in the History tab to download all your workout data as a CSV file. Open it in Excel, Google Sheets, or any spreadsheet app for custom analysis. Your data is always yours.

> 📊 **WHAT'S INCLUDED IN THE EXPORT**
> Date, workout duration, exercise name, set number, weight, reps, RPE, RIR, and any notes you logged.

14 ⚖️ BODY WEIGHT TRACKING

Log your body weight over time to track long-term trends alongside your training data.

LOGGING BODY WEIGHT

Go to the Progress tab and scroll to the Body Weight section. Enter today's weight and tap Save. A running chart shows your trend over time.

> 💡 **BEST PRACTICE**
>
> Weigh yourself first thing in the morning, after using the bathroom, and before eating or drinking. Consistency in conditions makes the trend line meaningful.

15 🔥 STREAK TRACKING

Your streak counts consecutive days with at least one completed workout. It's displayed prominently on the Home tab to keep the habit alive.

HOW STREAKS WORK

📅 Daily counting

Log at least one workout per day to extend your streak. Missing a day resets it to zero.

🔥 Streak display

Your current streak is shown as an orange badge on the Home tab. The number updates immediately after finishing a workout.

> 💡 **REST DAYS ARE PART OF THE PLAN.**
>
> A streak tracks training frequency — even a short 15-minute session counts. Active recovery work counts too.

16 👤 PROFILE & SETTINGS

The Profile tab is your personal hub — view and edit your info, set your training goal, review lifetime stats, and access app settings.

PROFILE FIELDS

FIELD	NOTES
First / Last Name	Shown as a greeting on the Home tab
Age	Optional — used for context in future coaching features
Weight	Your current body weight (lbs)
Height	Feet and inches
Country / City	Optional location fields
Goal	Build Muscle, Strength, Cardio, Cut / Lean Out, or Maintain

TRAINING GOALS

💪 **Build Muscle**
Hypertrophy & size

🏋️ **Strength**
Max power & 1RM

🏃 **Cardio**
Endurance & fitness

🔥 **Cut / Lean Out**
Fat loss & definition

⚖️ **Maintain**
Stay consistent

SETTINGS

Tap **Settings** on the Profile page to access:

🎨 **Theme**

Toggle between Dark mode and Light mode. Your preference is saved locally.

17 ☁ CLOUD SYNC & EXPORT

Your data is stored in Firebase Firestore and synced in real time across all your devices — no manual backup required.

HOW SYNC WORKS

⚡ **Real-time sync**

Every workout, profile change, and template is saved instantly to the cloud. Sign in on any device and your full history is there immediately.

📱 **Multiple devices**

Use Barbell Labs on your phone, tablet, and desktop simultaneously. All devices stay in sync automatically.

CSV EXPORT

To export your data, go to the **History tab** → **Export CSV**. The file includes every set from every workout ever logged. No lock-in — your data is always portable.

> 💡 DATA OWNERSHIP
>
> Barbell Labs never sells or shares your data. You own it completely. Export anytime and take it anywhere.

18 🔒 ACCOUNT & SECURITY

Manage your account credentials from the Profile tab → Settings → Security.

CHANGING YOUR EMAIL

1 **Go to Profile → Settings → Security → Change Email**

Enter your current password to verify your identity.

2 **Enter your new email address**

A verification link is sent to the new address.

3 **Verify the new address**

Click the link in the email. Your account email updates immediately.

CHANGING YOUR PASSWORD

Profile → Settings → Security → Change Password. Enter your current password, then your new password (same requirements as signup: min. 8 chars, 1 uppercase, 1 lowercase, 1 digit).

SIGNING OUT

A Sign Out button is always visible at the bottom of the Profile tab. Tapping it signs you out and returns you to the login screen. Your data remains safely stored in the cloud.

SIGN IN WITH GOOGLE

If you created your account with Google, your password is managed through your Google account. Email/password changes are not available for Google accounts.

🔒 SECURITY

> All data is encrypted in transit and at rest via Firebase. Barbell Labs never stores your password — it's handled entirely by Firebase Authentication.

BARBELLLABS

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